

Harini Vinu

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EDUCATION

Master of Science, Computer Science

New York University, New York

September 2024 — May 2026

GPA: 3.75/4.00

Courses: Data Structures & Algorithms, Distributed Systems, Machine Learning, Deep Learning, Big Data, Cloud Computing, Blockchain

Bachelor of Technology, Computer Science

SRM Institute of Science and Technology

September 2020 — May 2024

GPA: 3.8/4.00

Courses: Artificial Intelligence, OOP, Operating Systems, Computer Networks, System Design, Database Systems, Software Engineering

EXPERIENCE

Software Engineering Intern

Qualitest – CoCo (AI-driven Test Optimization Platform)

May 2025 — August 2025

New York, United States

- Engineered scalable backend services for AI-driven test orchestration using **Python, Redis, and MongoDB**, supporting distributed queue-based workflows.
- Implemented distributed job scheduling to coordinate **100+** concurrent tasks across automated testing workflows.
- Optimized large-scale data pipelines using asynchronous queue processing and distributed systems, reducing execution time by **40%** for high-throughput workloads.
- Improved distributed job reliability with monitoring, fault-tolerant retries, and failure recovery, reducing job failures by 30%.

Software Engineering Intern

LOCOMeX

June 2025 — August 2025

New York, United States

- Built scalable data validation and processing services handling **100K+** records, improving data consistency across large-scale software ingestion workflows.
- Improved **REST API** and query performance by optimizing hierarchical data models and database access patterns.
- Developed scalable microservices using event-driven architecture to propagate risk signals across supplier networks in real time, supporting low-latency distributed software workflows.
- Implemented aggregation and normalization for supplier data, improving ESG scoring and reducing latency by 30%.

Big Data Engineer Intern

New York University

January 2025 — May 2025

New York, United States

- Engineered distributed data pipelines using **Spark, Kafka, and Hadoop** to process **10M+** records across scalable computing clusters.
- Utilized AI productivity and automation tools to streamline development and troubleshooting, cutting issue-resolution time by **90%**.
- Optimized scalable data ingestion and processing modules with real-world datasets, improving distributed system efficiency by **25%**.
- Designed real-time streaming data ingestion systems using Kafka producers and consumers, enabling near-real-time analytics and reducing data latency for downstream reports.

Software Engineering Intern

Skill Vertex

December 2021 — February 2022

Chennai, India

- Developed a real-time hand detection and gesture recognition system using Python, OpenCV, and Google AI tools, improving detection accuracy by **40%**.
- Enhanced object tracking and optimized machine-learning pipelines using preprocessing and hyperparameter tuning, which cut false detections by **15%**, lowered latency by **25%**, and achieved real-time processing at 30 FPS on over 1,000 test samples.

PROJECTS

Dining Concierge Chatbot | AWS Lambda, API Gateway, DynamoDB [GitHub](#)

September 2025 — October 2025

- Built a serverless conversational application using **AWS Lambda, API Gateway, Lex, DynamoDB, Elasticsearch**, and the Yelp API to deliver real-time restaurant recommendations. Orchestrated asynchronous, event-driven workflows using **SQS** and **SES** for personalized notifications, with **CloudWatch** monitoring and logging to improve reliability and maintainability.

Real-Time Stock Market Analysis System | Kafka, Spark Streaming, Hadoop [GitHub](#)

September 2024 — October 2024

- Designed a distributed streaming analytics platform using **Kafka, Spark Streaming, & Hadoop** to process live financial data feeds. Built backend services for computing technical indicators and storing high-volume time-series data in **PostgreSQL & InfluxDB**, enabling low-latency analytics. Deployed data ingestion pipelines & used **Grafana** for performance monitoring & analytics visualization.

Real-Time Video Monitoring and Alert System | AWS Kinesis, Lambda, DynamoDB [GitHub](#)

January 2025 — May 2025

- Built a scalable real-time monitoring system using **AWS Kinesis, Lambda, SQS, DynamoDB, and Rekognition** to process live data streams and perform real-time image analysis, generating low-latency insights and alerts. Designed event-driven backend services to trigger notifications with millisecond-level latency and data storage layers for real-time monitoring dashboards. Implemented analytics pipelines using **Elasticsearch** and **AWS QuickSight** to analyze streaming event data and provide operational insights.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript, Go, Scala, Bash

Backend & APIs: Microservices Architecture, Spring Boot, FastAPI, Django, Flask, gRPC, GraphQL, REST APIs, OpenAPI, Swagger

Data & Machine Learning: TensorFlow, Scikit-learn, OpenCV, Computer Vision, ML Pipelines, Hadoop, Hive, Akka

Databases & Caching: MySQL, PostgreSQL, MongoDB, Redis, InfluxDB, SQLite, Hibernate

Cloud, DevOps, Distributed Systems & Monitoring: AWS (Kinesis, Step Functions, CloudFront, CloudFormation, Cognito, Rekognition), Serverless, Docker, Kubernetes, Terraform, Jenkins, Prometheus, Grafana, Splunk, Ansible, Linux/Unix

Certifications: NYU: CareerHub Data Science Bootcamp ; Accenture: AI; Infosys SpringBoard: Java Certified Foundation Associate